

Arduino Code:

```
// Motor control pins into the arduino
#define STEP_PIN 4
#define DIR_PIN 3

// Input pins into the arduino
#define LATCHING_BUTTON_PIN 5
#define LIMIT_SWITCH_PIN 6

// LED pins - input pins into the arduino
#define LED_RED_PIN 8
#define LED_GREEN_PIN 9
#define LED_BLUE_PIN 10

//extra credit input pins into the arduino (non-latching button inputs)
#define CCW_PIN 11 //clockwise and counterclockwise
#define CW_PIN 12

float largeSyringeDiameter = 18.7; //measured internal diameter of syringe
float smallSyringeDiameter = 14.6;

float flowRate = 5; //mL/min [EDIT THIS VALUE to change pump speed]
float syringeDiameter = smallSyringeDiameter; //(smallSyringeDiameter for 10 mL,
largeSyringeDiameter for 20 mL)
float distancePerRevolution = 8.0; //lead screw lead
float stepsPerRevolution = 3200; //3200 (16 steps) microstepping

#include <AccelStepper.h>

AccelStepper stepper(AccelStepper::DRIVER, 4 , 3); //used the STEP and DIR PIN input values
```

```

void setup() //initial set up: configuring pins to input_pullup/output
{
  stepper.setMaxSpeed(1000); //loop only runs once, so this just initializing values
  calculateStepperSpeed();
  pinMode(LED_RED_PIN, OUTPUT);
  pinMode(LED_GREEN_PIN, OUTPUT); //LED's outputs, buttons are INPUT_pullup
  pinMode(LED_BLUE_PIN, OUTPUT);
  pinMode(LATCHING_BUTTON_PIN, INPUT_PULLUP);
  pinMode(LIMIT_SWITCH_PIN, INPUT_PULLUP);
  setColor(0, 255, 0); //green initially
  pinMode(CCW_PIN, INPUT_PULLUP);
  pinMode(CW_PIN, INPUT_PULLUP);
}

void loop() //runs infinitely, constantly checking for changes in button states
{

  if (digitalRead(LATCHING_BUTTON_PIN) == LOW &&
  digitalRead(LIMIT_SWITCH_PIN) == HIGH){ //if latching button is pressed, run the pump
    calculateStepperSpeed();
    stepper.runSpeed();
    setColor(0, 255, 0); // Green Color
  }
  if ((digitalRead(LATCHING_BUTTON_PIN) == HIGH && digitalRead(LIMIT_SWITCH_PIN) ==
  HIGH)){ //if latching button is pressed again, pause the pump
    setColor(255, 50, 0); // Yellow Color
    if (digitalRead(CCW_PIN) == LOW){ //if its paused and the non-latching (1) is pressed
      setColor(255, 0, 255); //purple CCW //reverse for easy reset
      stepper.setSpeed(900);
      stepper.runSpeed();
    }
    if (digitalRead(CW_PIN) == LOW){ //same thing for the other non-latching (2)
      setColor(0, 0, 255); //blue CW //forward if reversed too far
      stepper.setSpeed(-900); // neg bc this is the forward direction
      stepper.runSpeed();
    }
  }
}

```

```

if (digitalRead(LATCHING_BUTTON_PIN) == LOW &&
digitalRead(LIMIT_SWITCH_PIN)==LOW){ //If the limit switch is triggered, stop the pump, it's
empty
    setColor(255, 0, 0); // red Color
}

if (digitalRead(LATCHING_BUTTON_PIN) == HIGH && digitalRead(CCW_PIN) == LOW){
//pause after limit switch red, then can reverse
if (digitalRead(CCW_PIN) == LOW){
    stepper.setSpeed(900);
    stepper.runSpeed();
}

}

}

void setColor(int redValue, int greenValue, int blueValue) {
analogWrite(LED_RED_PIN, redValue); //simple set color function to make setting RGB values
easier in void loop
analogWrite(LED_GREEN_PIN, greenValue);
analogWrite(LED_BLUE_PIN, blueValue);
}

void calculateStepperSpeed() { //calculating stepper speed (see calculation section for
explanation)
float syringeArea = 3.14159 * pow(syringeDiameter / 2.0, 2);
float flowRateMmCubed = flowRate * 1000.0;
float linearSpeed = flowRateMmCubed / syringeArea;
float stepperSpeed = (linearSpeed / distancePerRevolution) * stepsPerRevolution / 60.0; //steps
per second
stepper.setSpeed(-stepperSpeed); //neg bc this is the forward direction

}

```